Customer Churn Analysis & Prediction

Final Business Report - Telecommunications Customer Retention

Dataset: IBM Telco Customer Churn sample (7,043 customers, 21 columns). All figures in this report are produced by the accompanying notebook (*notebooks/Customer_Churn_Analysis.ipynb*).

1. Executive summary

- **The problem:** 1,869 of 7,043 customers (**26.5%**) churned. They carried **$139,131 per month** of recurring revenue, 30.5% of the monthly total in the file.
- **Where churn concentrates:** month-to-month contracts (42.7% churn; 88.6% of all churners), customers in their first 12 months (47.4%), fiber-optic internet (41.9%; 54.6% when also month-to-month), electronic-check payers (45.3%), and internet customers without security or tech-support add-ons (about 42%). Two-year contracts churn at only 2.8%.
- **The model:** six algorithms were compared under identical conditions. All reach ROC-AUC 0.83-0.85, well above the 0.50 of guessing; the top three are statistically tied. **Logistic Regression was selected** for stability, transparency and speed (test ROC-AUC 0.841; it catches 78% of churners at the default setting and 90% at a 0.35 threshold).
- **Practical use:** the riskiest 20% of customers, as ranked by the model, contain 50% of all churners (66% of that group actually churn). The model should be used to prioritise human-led retention efforts, not to make automatic decisions.
- **Key caution:** these are associations in a historical sample. They do not prove that changing a contract, price or service would change customer behaviour; every action below should be piloted with a control group.

2. Objective, data and method

Objective. Understand patterns associated with churn, evaluate classification models, and translate supported findings into retention recommendations.

Data checks. The raw file was inspected before any change: 7,043 rows, 21 columns, no duplicates, no NaN values. One hidden issue was found: **TotalCharges is stored as text and contains 11 blank strings**. All 11 belong to customers with tenure = 0 (new customers not yet billed), none churned. They were set to 0 and kept rather than imputed with an average, which would invent charges that never occurred. No outliers were detected (IQR rule) and all categories were consistent. The raw file was never modified.

Method. One engineered feature (number of internet add-ons). Stratified 80/20 train/test split (5,634 / 1,409; churn 26.5% in both). Scaling and one-hot encoding inside scikit-learn Pipelines so no test information leaks into training. Six models (Logistic Regression, Decision Tree, Random Forest, KNN, SVM, Gradient Boosting), each tuned with 5-fold stratified cross-validation on ROC-AUC using training data only; a majority-class baseline for reference. Class weights were used where supported because the target is moderately imbalanced.

3. What the data shows

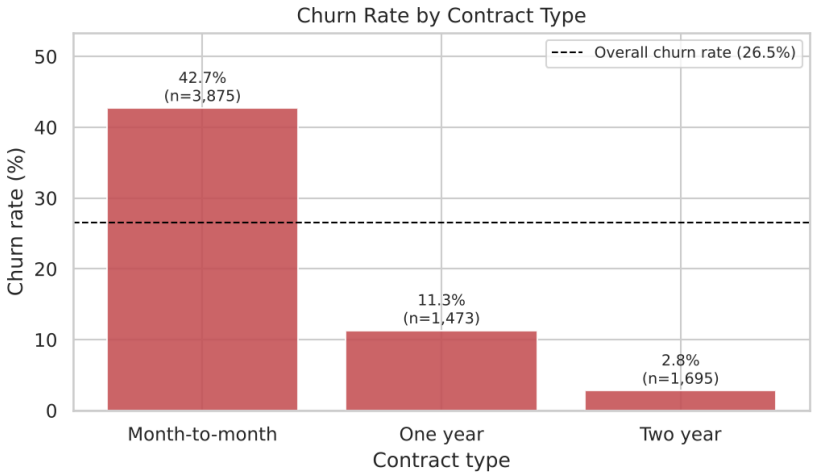


Figure 1. Churn rate by contract type. Month-to-month customers churn about 15 times as often as two-year customers.

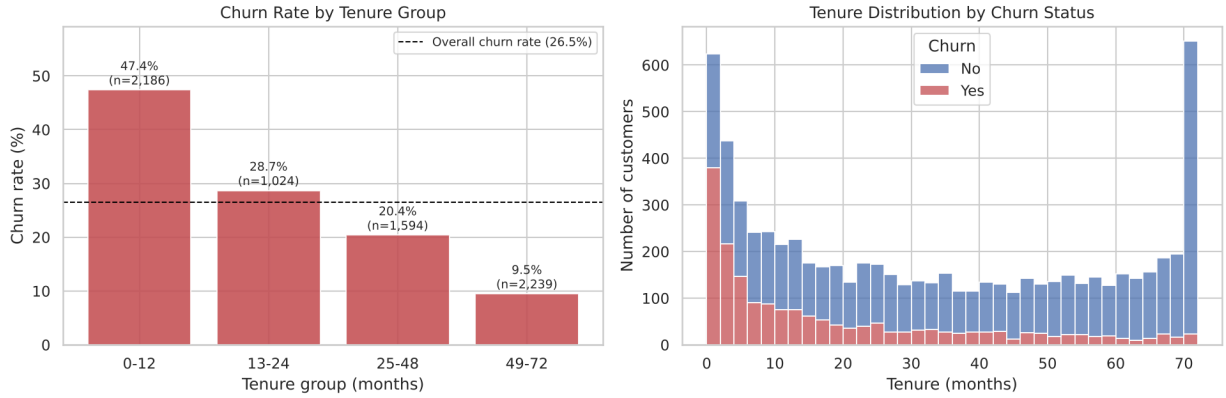


Figure 2. Churn is highest in the first year (47.4%) and falls to 9.5% after four years.

Factor	Group	Churn rate	Customers
Contract	Month-to-month / One year / Two year	42.7% / 11.3% / 2.8%	3,875 / 1,473 / 1,695
Tenure	0-12 / 13-24 / 25-48 / 49-72 months	47.4% / 28.7% / 20.4% / 9.5%	2,186 / 1,024 / 1,594 / 2,239
Internet	Fiber / DSL / None	41.9% / 19.0% / 7.4%	3,096 / 2,421 / 1,526
Payment	Electronic check / Mailed / Bank transfer (auto) / Credit card (auto)	45.3% / 19.1% / 16.7% / 15.2%	2,365 / 1,612 / 1,544 / 1,522
Online Security*	No / Yes	41.8% / 14.6%	3,498 / 2,019
Tech Support*	No / Yes	41.6% / 15.2%	3,473 / 2,044
Senior citizen	Yes / No	41.7% / 23.6%	1,142 / 5,901
Gender	Female / Male	26.9% / 26.2%	3,488 / 3,555

*Internet customers only (the 1,526 customers without internet service are excluded from these two rows).

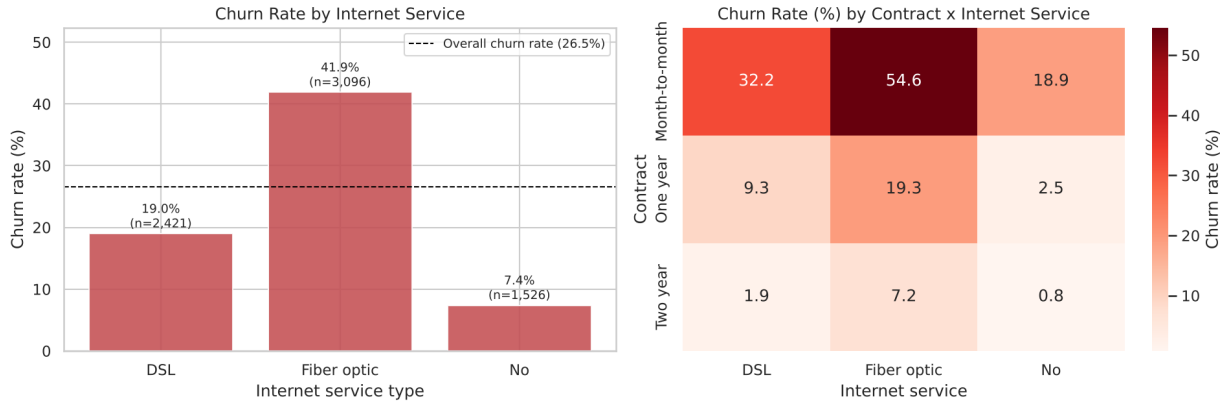


Figure 3. Fiber customers churn more than DSL customers within every contract type; month-to-month fiber reaches 54.6%.

Interpretation. Several of these factors overlap (fiber customers are more often month-to-month and pay by electronic check), so raw gaps overstate what any single factor does. Still, the fiber gap persists inside every contract type (month-to-month 54.6% vs 32.2% for DSL; one-year 19.3% vs 9.3%; two-year 7.2% vs 1.9%). Gender shows no meaningful difference. Streaming add-ons differ by only 3-4 points, whereas security and support add-ons differ by 26-27 points.

4. Model comparison

Model	Test ROC-AUC	CV ROC-AUC (std)	Recall	Precision	F1	Accuracy
Gradient Boosting	0.846	0.850 (0.012)	0.503	0.660	0.571	0.799
Random Forest	0.844	0.848 (0.010)	0.791	0.517	0.625	0.748
Logistic Regression (selected)	**0.841**	**0.846 (0.013)**	**0.783**	**0.504**	**0.614**	**0.738**
Decision Tree	0.836	0.833 (0.013)	0.781	0.505	0.613	0.739
SVM (RBF)	0.834	0.842 (0.011)	0.626	0.594	0.609	0.787
KNN	0.831	0.837 (0.012)	0.575	0.592	0.583	0.782
Baseline (always 'no churn')	0.500	-	0.000	-	0.000	0.735

Test set: 1,409 customers (374 churners). Recall, precision, F1 and accuracy use a 0.50 probability threshold.

Why Logistic Regression. (1) The top three models are not distinguishable on ranking quality: the 0.004 CV-AUC gap between Gradient Boosting and Logistic Regression is smaller than the fold-to-fold standard deviation (about 0.01-0.013). (2) Logistic Regression has the smallest overfitting gap between training and cross-validated AUC (0.003, versus 0.031 for Random Forest). (3) It is transparent and auditable, and trains in about one second. (4) It already catches 78% of churners at the default setting.

Accuracy is not the selection criterion. The models with the highest accuracy (Gradient Boosting 0.799, SVM 0.787, KNN 0.782) miss the most churners (recall 0.50-0.63), and a model that always predicts 'no churn' already scores 0.735 accuracy. **Trade-off:** Gradient Boosting and Random Forest have slightly higher PR-AUC (0.664 and 0.656 vs 0.627); with only 374 test churners this is not conclusive and is worth re-testing on newer data.

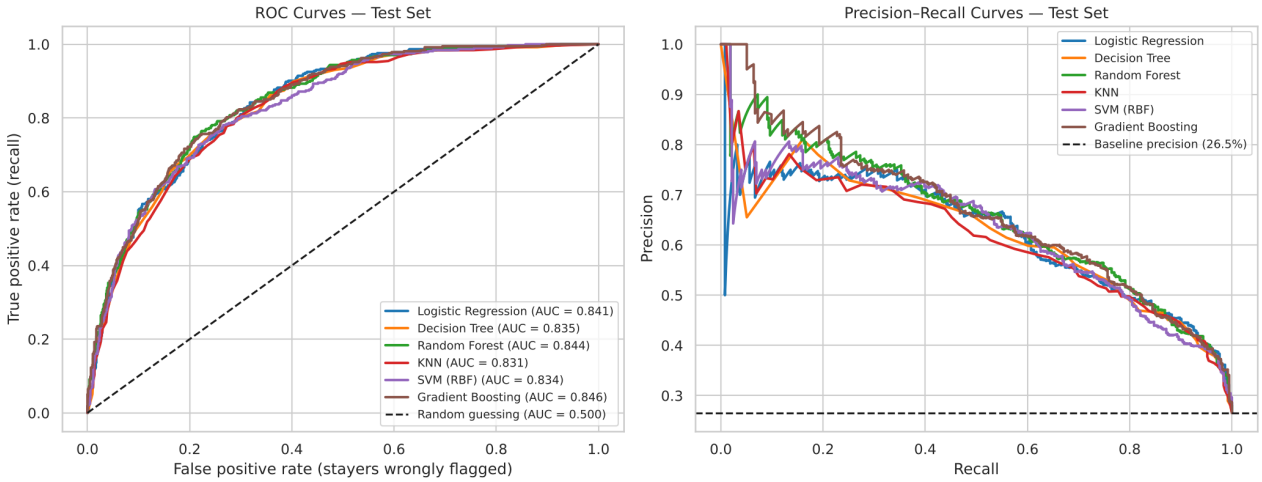


Figure 4. ROC and precision-recall curves for all six models on the test set.

5. Choosing a threshold and using the model

The model outputs a risk score; the business chooses the cut-off. Lowering it catches more churners but flags more customers who would have stayed. The threshold was chosen on out-of-fold *training* predictions (maximising F2, which weights recall twice as much as precision, under the assumption that missing a churner costs more than a wasted offer), then evaluated once on the test set.

Threshold	Customers flagged	Churners caught (of 374)	Churners missed	False alarms	Precision	Recall
0.50 (default)	41.2%	293	81	288	0.504	0.783
0.35 (chosen)	53.8%	337	37	421	0.445	0.901

Contact the riskiest...	Customers	% of all churners reached	Share of group who churn
10%	141	28.3%	75.2%
20%	282	50.0%	66.3%
30%	423	64.4%	57.0%
40%	564	77.3%	51.2%
50%	704	87.2%	46.3%

Flagging 54% of customers is probably too many for a costly offer. We therefore recommend tiers: high-touch contact for the top ~20%, low-cost automated outreach further down. The final cut-off should be set from the cost of an offer versus the value of a retained customer; this data does not contain those figures, so 0.35 is a starting assumption, not a verified optimum.

6. What drives risk in the model

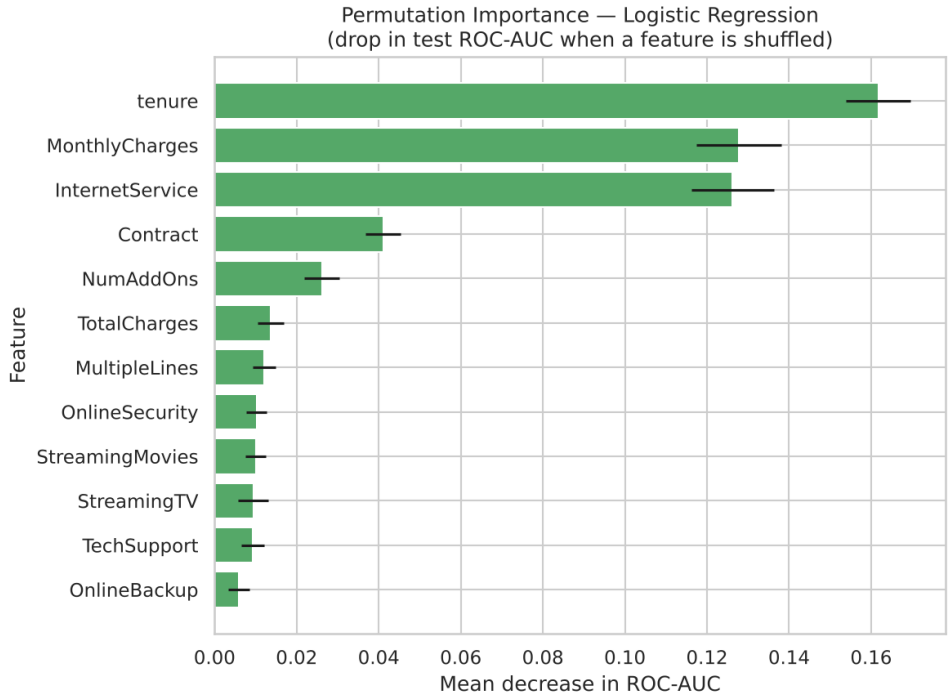


Figure 5. Permutation importance: drop in test ROC-AUC when each feature is shuffled.

The most influential features are **tenure** (AUC drop 0.162), **monthly charges** (0.128) and **internet service type** (0.126), followed by contract (0.041) and number of add-ons (0.026). Demographics and payment method do not appear among the top 12: their raw associations are largely captured by contract, tenure and service type, so electronic check looks more like a marker of risky customers than an independent driver. Some individual regression coefficients (for example monthly charges) look counter-intuitive because price is almost fully determined by the services chosen (multicollinearity); they should not be read one by one. **Plain language:** customers who are new, on month-to-month contracts, on fiber, paying higher bills and without security or support add-ons are the most likely to leave.

7. Recommendations

#	Recommendation	Evidence in this dataset
1	**Focus on the first year for month-to-month customers**: onboarding check-ins and early-life offers.	Month-to-month customers in months 0-12: 1,994 customers, 51.4% churn, 54.8% of all churners.
2	**Encourage 1- or 2-year contracts** with a modest benefit; pilot with a control group (loyal customers may self-select).	Churn 42.7% / 11.3% / 2.8% by contract; month-to-month = 88.6% of churners.
3	**Investigate the fiber proposition** (quality, price vs competitors, exit survey) before discounting.	Fiber 41.9% vs DSL 19.0%. Month-to-month fiber: 2,128 customers, 54.6% churn, $100,482/month lost.
4	**Trial security / tech-support bundles** for at-risk internet customers; skip streaming promotions.	Without security 41.8% vs 14.6%; internet customers with neither security nor support: 2,553 customers, 49.0% churn, 66.9% of churners. Churn falls from 52.2% (0 add-ons) to 5.3% (6).
5	**Move electronic-check payers to automatic payment** with a small incentive; use electronic check as a risk flag.	45.3% churn (2,365 customers) vs 15.2-16.7% for automatic methods. Expect a smaller effect than the raw gap, as the model finds little extra signal in payment method.
6	**Review pricing for high-bill, no-commitment customers** (price lock or plan right-sizing in exchange for a contract).	Churner median bill $79.65 vs $64.43; churn 37.8% in the $70-90 band vs 9.8% at $30 or less. Price cannot be separated from service type here.
7	**Run a model-guided tiered campaign** with a random hold-out group to measure whether offers work.	Top 20% by risk reaches 50% of churners at 66% precision. The model predicts who may leave, not who will respond.
8	**Support senior customers with tailored service** (not worse terms); complete a fairness review first.	1,142 seniors (16%), 41.7% churn vs 23.6%.

No saving or churn-reduction figure is claimed: that requires pilot results. Gender-based campaigns are not supported by the data (26.9% vs 26.2%).

8. Limitations and next steps

- **Associations, not causes:** the data cannot show that changing contract, price or add-ons would change behaviour.
- **Snapshot without dates:** a churner's tenure stops when they leave, so part of the 'short tenure' signal is built into how the data was recorded. Performance on a true 'who leaves next month' task may be lower than reported.
- **Missing drivers:** no reasons for leaving, usage, network quality, complaints, competitor prices or offer history.
- **One sample, one split:** 374 test churners; differences of about 0.01 AUC are within noise. Results may not carry over to the current customer base.
- **Scores are not calibrated probabilities** (class weights were used) - use them for ranking and thresholding.
- **Fairness not audited:** SeniorCitizen and gender are model inputs. Review before any deployment.
- **Out of scope:** real-time scoring, CRM integration, automated outreach and any guarantee of churn reduction.
- **Data that would improve the analysis:** churn dates and reasons, usage and network-quality data, support history, competitor pricing, previous offers and responses, customer margin and acquisition cost, and time-stamped snapshots.
- **Next steps:** confirm cost assumptions to set the threshold; pilot recommendations 1-7 with control groups; re-test the model on recent data; complete a fairness review.

Source: IBM Telco Customer Churn sample dataset (github.com/IBM/telco-customer-churn-on-icp4d; Kaggle mirror: kaggle.com/datasets/blastchar/telco-customer-churn). Confirm dataset terms before redistribution or commercial use.